

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 1

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Express $2x^2 - 12x + 7$ in the form $a(x + b)^2 + c$, where a , b and c are constants. Hence state the minimum value of $2x^2 - 12x + 7$ and the value of x at which it occurs. **[3]**

2 Solve the equation $2 \sin^2 \theta + 3 \cos \theta = 3$ for $0^\circ \leq \theta \leq 360^\circ$. **[4]**

3 The points A and B have coordinates $(1, 5)$ and $(7, -3)$ respectively. Find the equation of the perpendicular bisector of AB , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers. **[4]**

4 A geometric progression has first term a and common ratio r . The second term is 12 and the sum to infinity is 64.

(a) Show that $16r^2 - 16r + 3 = 0$ and hence find the two possible values of r . [3]

(b) For the progression with the larger value of r , find the sum of the first 5 terms. [2]

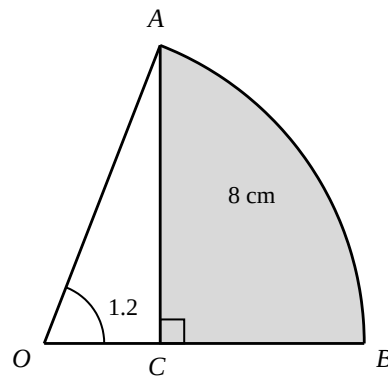
5 Functions f and g are defined by

$$f(x) = 2x + 3 \quad \text{for } x \in \mathbb{R}, \quad g(x) = x^2 - 4x \quad \text{for } x \geq 2.$$

(a) Find an expression for $fg(x)$ and solve the equation $fg(x) = 13$. [3]

(b) Find an expression for $g^{-1}(x)$ and state the domain of g^{-1} . [3]

6



The diagram shows a sector OAB of a circle with centre O and radius 8 cm. Angle $AOB = 1.2$ radians. The point C lies on OB and AC is perpendicular to OB .

(a) Find the perimeter of the shaded region ABC (bounded by the arc AB and the lines BC and CA). [3]

(b) Find the area of the region ABC . [3]

7 The equation of a curve is $y = x^3 - 6x^2 + 9x + 1$.

(a) Find the coordinates of the stationary points of the curve and determine the nature of each. [4]

(b) Find the equation of the normal to the curve at the point where $x = 2$, giving your answer in the form $ax + by + c = 0$. [3]

8 The line $y = x$ meets the curve $y = 4x - x^2$ at the origin O and at the point P .

(a) Find the coordinates of P . [2]

(b) Find the area of the region enclosed by the line and the curve. [4]

(c) The region bounded by the curve $y = 4x - x^2$, the x -axis and the line $x = 3$ is rotated through 360° about the x -axis. Find the volume of the solid obtained, giving your answer in terms of π . [3]

9

(a) Prove the identity $\frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta} \equiv \frac{2 \tan \theta}{\cos \theta}$. [3]

(b) Hence solve the equation $\frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta} = 8 \tan \theta$ for $0^\circ < \theta < 360^\circ$. [4]

(c) Solve the equation $\sin\left(2x - \frac{\pi}{3}\right) = \frac{1}{2}$ for $0 \leq x \leq \pi$, giving your answers in terms of π . [3]

10 A circle has equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

(a) Find the coordinates of the centre and the radius of the circle. [2]

(b) The line $y = 2x + k$ is a tangent to the circle. Find the possible values of the constant k , giving your answers in exact form. [4]

(c) Show that the point $P(7, 1)$ lies on the circle and find the equation of the tangent to the circle at P , giving your answer in the form $ax + by + c = 0$. [4]

11 The 5th term of an arithmetic progression is 17 and the sum of the first 10 terms is 185.

(a) Find the first term and the common difference of the progression. [3]

(b) Find the least value of n for which the sum of the first n terms exceeds 1000. [3]

12

(a) The curve $y = x^2 - 4x + 1$ is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Find the equation of the resulting curve, giving your answer in the form $y = ax^2 + bx + c$. [2]

(b) Describe fully a sequence of two transformations which maps the graph of $y = \cos x$ onto the graph of $y = 3 \cos 2x$. [3]

END OF PAPER (Total: 75 marks)